

Literature Discovery and Understanding

AI for Research: From Literature Discovery to Academic Writing · UTM Big Data Centre

● KEY TAKEAWAYS

- A good Boolean search string groups synonyms with OR and connects concepts with AND.
- Start with core multidisciplinary databases, then widen with AI-powered discovery tools.
- Citation metrics (Publish or Perish) rank impact, but never replace reading the paper.
- Always verify a Scopus AI synthesis against the original, cited publications.

● BOOLEAN SEARCH STRING PATTERN

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("[synonym 1]" OR "[synonym 2]") AND ("[synonym 1]" OR "[synonym 2]") AND "[concept 3]"
```

Adjust field tags per database: TITLE-ABS-KEY(...) in Scopus, TS= (...) in Web of Science.

● GLOSSARY

Boolean operators

AND, OR and NOT, used to combine or exclude search terms.

h-index

A metric measuring both productivity and citation impact of an author's publications.

Semantic search

Search based on meaning and context rather than exact keyword matches.



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drshahizan.github.io/bdc/module-1-literature-discovery.html



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Idea Generation and Research Design

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● KEY TAKEAWAYS

- A strong research question is Feasible, Interesting, Novel, Ethical and Relevant (FINER).
- AI can propose and assess candidate questions quickly, but you decide which one to pursue.
- Check novelty with tools like Scite, Consensus or Litmaps before assuming a gap is real.
- Gemini Notebook only reasons over what you upload, and it cannot see the wider literature.

● RESEARCH QUESTION PROMPT PATTERN

Act as an experienced research supervisor in [field].

My broad topic or interest area: [topic].

Target population or context: [population/context].

Suggest 3 to 5 candidate research questions and assess each one against FINER (Feasible, Interesting, Novel, Ethical, Relevant).

● GLOSSARY

FINER

Feasible, Interesting, Novel, Ethical and Relevant, a checklist for evaluating research questions.

Research gap

An unanswered question or unresolved contradiction in the existing literature.

Mixed-methods

A research design combining quantitative and qualitative data collection and analysis.



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Knowledge Organisation and Mind Mapping

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● KEY TAKEAWAYS

- A synthesis matrix, timeline, mind map or framework map each suit a different organisation goal.
- Always ground AI-generated structure in your actual sources; never let it invent connections.
- Mind-mapping tools structure your thinking; Gemini Notebook Studio generates a map from verified sources.
- A well-organised literature map becomes the backbone of your literature review section later.

● KNOWLEDGE ORGANISATION PROMPT PATTERN

Act as a research assistant helping me organise source material.

Sources, notes or topic to organise: [list your sources].
Organisation goal: [thematic synthesis matrix / timeline / mind map].

Ground every theme or connection in the sources provided. Do not invent themes or relationships that are not supported by the material.

● GLOSSARY

Synthesis matrix

A table organising sources by theme, showing which studies support each theme.

Literature matrix

A structured table comparing sources across dimensions such as method and findings.

Node

A single concept or entry in a mind map, connected to related ideas by branches.



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Data and Research Visualisation

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● KEY TAKEAWAYS

- Match the chart type to your goal, rather than choosing a chart first and forcing data to fit it.
- Bibliometric tools (VOSviewer, CiteSpace, Gephi, Bibliometrix) reveal patterns across a whole literature.
- Keep axes honest and labels clear. A chart should clarify data, not distort it.
- Studio prompts can generate a source-grounded visual directly from your verified papers.

● CHART-TYPE DECISION PATTERN

Goal: [comparison / trend / proportion / relationship / distribution / process / geography]

Variables: [number of variables]

Audience: [academic or technical / general audience]

Ask: which chart type shows this comparison most honestly, without distorting the underlying data?

● GLOSSARY

Co-citation network

A map showing which papers are frequently cited together.

Bibliometric analysis

The quantitative study of publication and citation patterns.

Data storytelling

Presenting data with a narrative and clear takeaway, not just raw numbers.



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Academic Writing and Deep Research Assistance

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● KEY TAKEAWAYS

- Use AI to polish clarity and structure, but keep your own argument, voice and judgement in control.
- Verify every fact, statistic, quotation and citation an AI drafted or paraphrased.
- Disclose AI use where your journal or institution requires it.
- A citation manager (Zotero, Mendeley or EndNote) removes formatting errors so you can focus on the argument.

● AI USE DISCLOSURE PATTERN

During the preparation of this manuscript, the author(s) used [tool name] to [purpose, e.g. improve clarity and readability] and [tool name] to [purpose, e.g. assist in identifying related literature].

All AI-suggested content was reviewed, verified against original sources, and edited by the author(s), who take full responsibility for the content of this publication.

● GLOSSARY

DOI

Digital Object Identifier, a permanent link to a specific published work.

Deep research

An AI feature that searches, reads and synthesises multiple sources into one report.

AI disclosure

A statement in a manuscript declaring where and how AI tools were used.



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